

Guidelines for Using Generative AI in Sho's Courses (July 2026)

Generative AIs (GAIs) are powerful tools that assist us in many tasks^{#1#2}. Improving greatly in recent years, they were merely chatbots in 2024, but now agentic systems can autonomously plan, write, and debug software, or operate applications such as web browsers to accomplish complex tasks.

As of 2026, free-to-use GAIs outperform most university students in physics and mathematics. Employers have little reason to hire someone who cannot use GAI effectively. The ability to work with GAI is now a baseline expectation for university graduates.

At the same time, a solid foundation of your own is required to use GAIs appropriately. You need the basics in your head, ready to use without looking them up. Otherwise, you cannot judge whether a GAI answer is right or wrong. If you rely on GAI without understanding, you become a human-shaped interface that merely relays GAI output to others. Through your university education, you should *learn how to use GAIs effectively and build a solid foundation of knowledge and judgment for their responsible use*.

This guideline is provided to maximize your academic success and learning, and is based on Sho's opinions and personal understanding of modern GAIs. If you attend Sho's lecture courses, you should read this document together with [Sho's Rules and Suggestions for Assignments](#)^{#3}. Although it is impossible to detect GAI use with 100% confidence, Sho reserves the right to give reduced or negative scores when violations are suspected.

#1: Previous versions of this document are available on GitHub: https://github.com/misho104/LectureCommon/commits/main/Notes/generative_ai.pdf. You will notice how much the situation has changed due to the rapid improvement of GAIs.

#2: This document was prepared with the assistance of GAIs. However, this kind of remark is no longer meaningful since we always use GAIs.

#3: https://misho104.github.io/LecturePublic/submission_guideline.pdf

Regulations on Coding Assignments

For coding assignments, Sho **encourages** you to use GAIs, including agentic tools, and Sho evaluates your work assuming you have used them. Following the rules given in [the guideline](#), you must understand everything you write, including the codes. Sho may ask you to explain the overall structure of your code, why you chose a particular approach, and how any nontrivial component works.

Regulations on Other Assignments

For other assignments, Sho **does not forbid** your use of GAIs. As you need to follow [the guideline](#),

- If you use GAI tools, you **must** state so clearly by, for example, writing “*I used ChatGPT to solve this problem*”, or put the mark [AI used](#) on each relevant part of your submission.
- You **must** be able to explain everything in your submission.

The key question is whether your submission demonstrates your own effort and understanding.

Remarks on AI and Physics

As of July 2026, GAIs are highly reliable for coding and language-related tasks. In scientific and mathematical fields, the situation has also changed significantly. Recent *reasoning models* can solve

problems at the level of physics olympiads. They are no longer the “often wrong in science” tools of a few years ago.

This has two implications for you. First, GAI can be useful tutors and problem-solving assistants. However, they may make mistakes, especially on difficult or unfamiliar problems. Read their explanations carefully, and if you have doubts or do not fully understand them, ask Sho or other professors.

Second, and more importantly, if you can only solve the physics problems that GAI can solve, you will have no advantage over GAI. The goal of studying physics is not to compute answers, but to develop critical thinking: asking the right questions, recognizing mistakes in reasoning, and finding the best approach. These skills are also essential in the AI era, where you will be expected to use GAI responsibly.

Other guidelines

Guidelines on using GAI are available online, such as

- [Using Generative AI Reflectively and Responsibly in Teaching and Learning](#) (UCLA)
- [Generative AI for Students](#) (Open University)
- [AtCoder Rules against Generative AI](#)
- [AI ツールの授業における利用について](#) (the University of Tokyo; in Japanese)

However, due to the rapid development of GAI, documents written in 2022–2024 are now largely obsolete, and those written in 2025 must be read carefully. Check the last-update date.